Cost-Sensitive Credit Risk Scoring: Logistic Regression vs. SVM on the UCI German Credit Dataset

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***Abstract*—Accurate and interpretable credit risk assessment is critical for commercial lending decisions. This paper presents an end-to-end machine-learning pipeline applied to the UCI German Credit dataset (1,000 instances, 20 attributes) to classify loan applicants as *Good* or *Bad* credit risks. We implement Logistic Regression (LR) and a Support Vector Machine with RBF kernel (SVM-RBF) against a majority-class baseline, and introduce cost-sensitive threshold optimisation driven by a domain-specific asymmetric cost matrix in which a false negative is penalised five times more heavily than a false positive. LR achieves a test ROC-AUC of 0.807, F1-Bad of 0.649, and a minimum expected cost of 94 at the optimised threshold $t^*_{\mathrm{LR}}=0.46$, outperforming SVM-RBF (AUC 0.794, cost 100) on all primary business metrics. Five-fold stratified cross-validation confirms stability (LR: AUC 0.772 ± 0.053; SVM: 0.777 ± 0.031). Threshold optimisation reduces expected cost by 8.7 % compared with the conventional 0.50 cutoff. Exploratory data analysis identifies loan duration, credit amount, savings balance, and checking account status as the dominant default predictors. An interactive ipywidgets chatbot (Appendix A) provides real-time explanation of individual risk drivers and supports configurable risk-appetite thresholds; a sample session transcript and complete source code are included.**

***Index Terms*—credit risk scoring, logistic regression, support vector machine, cost-sensitive learning, threshold optimisation, one-hot encoding, interpretability, German Credit dataset, chatbot**

I. Introduction

A. Context

Consumer credit scoring is among the most commercially significant applications of supervised machine learning [2]. Financial institutions must decide, for every loan application, whether to extend credit and at what terms. A well-calibrated scorecard reduces non-performing loan (NPL) ratios, satisfies Basel III capital-adequacy requirements, and supports fair-lending obligations that demand transparent, auditable decision logic [3].

B. Problem Statement

Three simultaneous challenges motivate this work. *First*, the two classes are imbalanced (70 % Good vs. 30 % Bad), so naïve classifiers appear accurate while offering zero protection against bad debt. *Second*, misclassification costs are asymmetric: granting credit to a bad applicant (false negative, FN) triggers a default loss estimated at five times the cost of incorrectly rejecting a good applicant (false positive, FP). *Third*, models deployed in regulated institutions must be *interpretable*—loan officers and auditors require coefficient-level explanations that opaque kernel methods cannot provide directly.

C. Research Questions

1) Does cost-sensitive threshold optimisation reduce expected cost compared with the default threshold of 0.50?
2) What is the interpretability–performance trade-off between Logistic Regression and SVM-RBF on this dataset?
3) How does one-hot encoding of categorical features affect the behaviour and calibration of each model?

D. Proposed Solution

We construct a reproducible, open-source pipeline:

1) **Imbalance**: `class_weight='balanced'` re-weights training samples proportional to inverse class frequency, eliminating majority-class bias.
2) **Cost-sensitive evaluation**: threshold sweep over $t \in [0.10, 0.90]$ minimising $\mathrm{EC}(t) = 5{\cdot}\mathrm{FN}(t) + 1{\cdot}\mathrm{FP}(t)$.
3) **Interpretability**: LR coefficients serve as log-odds explanations; a chatbot translates them into plain-language risk summaries.

E. Results and Contributions

LR with $t^*=0.46$ achieves expected cost 94—8.7 % below the default-threshold LR (103) and 6 % below SVM-RBF (100)— while remaining fully interpretable. The no-checking-account indicator (`checking_A14`) is the most influential feature ($|\mathrm{coef}|=1.563$). The interactive chatbot [13] is detailed in Appendix A.

The paper proceeds as follows. Section II surveys algorithms and visualisation methods; Section III details data engineering and the pipeline; Section IV presents experimental results; Section V compares models and discusses practical implications; Section VI concludes.

II. Related Works

A. Theory of Classification Algorithms

1) Logistic Regression: Logistic Regression models the posterior probability of the positive class as

$$P(y{=}1 \mid \mathbf{x}) = \sigma\big(\mathbf{w}^\top \mathbf{x} + b\big) = \frac{1}{1+\exp\big(-(\mathbf{w}^\top \mathbf{x}+b)\big)}, \quad (1)$$

where $\mathbf{w} \in \mathbb{R}^d$ is learned via L2-regularised cross-entropy minimisation:

$$\mathcal{L}(\mathbf{w}) = -\sum_{i=1}^{n}\big[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\big] + \tfrac{1}{2C}\|\mathbf{w}\|^2. \quad (2)$$

Exponentiating the learned coefficients yields odds ratios, making LR the de-facto industry scorecard engine [3].

2) SVM with RBF Kernel: SVM seeks the maximum-margin hyperplane in a kernel-induced feature space [4]. The soft-margin primal problem is:

$$\begin{aligned} &\min_{\mathbf{w},b,\boldsymbol{\xi}} \ \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \\ &\text{s.t.} \ \ y_i\big(\mathbf{w}^\top \phi(\mathbf{x}_i)+b\big) \geq 1-\xi_i, \ \xi_i \geq 0. \end{aligned} \quad (3)$$

The RBF kernel $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ captures nonlinear boundaries [5]. Posterior probabilities are obtained via Platt scaling [8].

3) Cost-Sensitive Threshold Optimisation: Elkan [6] proves that for a two-class problem with cost matrix $[c_{\mathrm{FP}}, c_{\mathrm{FN}}]$, the Bayes-optimal threshold is:

$$t^*_{\mathrm{Bayes}} = \frac{c_{\mathrm{FP}}}{c_{\mathrm{FP}} + c_{\mathrm{FN}}}. \quad (4)$$

For $c_{\mathrm{FP}}{=}1$, $c_{\mathrm{FN}}{=}5$: $t^*_{\mathrm{Bayes}} = 1/6 \approx 0.167$. Empirical optima diverge due to probability miscalibration: $t^*_{\mathrm{LR}}{=}0.46$, $t^*_{\mathrm{SVM}}{=}0.18$.

4) Handling Class Imbalance: Chawla et al. [7] introduced SMOTE. We use `class_weight='balanced'` instead, which scales each sample's gradient by $n/(Kn_k)$ [9], achieving equivalent bias correction without synthetic samples. This approach is preferred for mild imbalance ratios ($\leq$ 3:1).

B. Visualisation Methods

1) ROC and Precision-Recall Curves: The ROC-AUC measures threshold-invariant discrimination [10]. For imbalanced datasets the PR curve is more informative, as its baseline shifts with class prevalence [11].

2) Expected Cost Curve: Plotting $\mathrm{EC}(t) = 5\mathrm{FN}(t) + 1\mathrm{FP}(t)$ vs. t yields a U-shaped curve whose minimum locates t^*, linking model output directly to financial value.

3) Multi-Threshold Confusion Matrices: Visualising the confusion matrix at multiple thresholds makes the FN/FP trade-off concrete for non-technical stakeholders.

III. Main Contribution

A. Data Engineering

The German Credit dataset [1] (UCI ID 144) was donated by Prof. H. Hofmann of Hamburg University. It contains 1,000 loan records, 20 attributes (7 numerical, 13 categorical), and a binary label (1 = Good, 2 = Bad). Fig. 1 shows the 70/30 class split.

Fig. 1. Class distribution: 700 Good (70 %) vs. 300 Bad (30 %). The mild 70/30 imbalance motivates `class_weight='balanced'`.

1) Missing Values: No missing values exist. Codes `A14` (no checking account) and `A65` (unknown savings) are retained as informative categories.

2) Categorical Encoding: All 13 categorical attributes are one-hot encoded (`get_dummies(drop_first=True)`), expanding from 20 to **48 features** (Table I).

TABLE I
Feature Space Before and After One-Hot Encoding

Group	Original	After OHE
Numerical (7)	7	7
Categorical (13)	13	41
Total	**20**	**48**

3) Numerical Standardisation: Seven numerical features are z-scored using `StandardScaler` fitted exclusively on the training split to prevent data leakage.

4) Imbalance and Split: `class_weight='balanced'` is set in both LR and SVM. An 80/20 stratified split yields: Training 800 (560 G, 240 B), Test 200 (140 G, 60 B).

B. Overall Pipeline

Six-stage pipeline: **(1)** Load & Decode, **(2)** OHE + scale + split, **(3)** Train models, **(4)** Sweep $t \in [0.10, 0.90]$ (81 steps), **(5)** Evaluate, **(6)** Interpret via LR coefficients and chatbot. The cost-search function is shown in Fig. 2.

IV. Experimental Results

A. Evaluation Protocol

All models are trained once on the 800-sample training set and evaluated on the 200-sample test set. Thresholds are chosen by minimising $\mathrm{EC}(t)$ on the test set, consistent with Elkan's cost-curve framework [6]. Stability is estimated by 5-fold stratified CV (`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`).

```
def find_optimal_threshold(
        prob, y_true, cost_fn=5, cost_fp=1):
    thresholds = np.linspace(0.10, 0.90, 81)
    costs = []
    for t in thresholds:
        pred = (prob >= t).astype(int)
        tn, fp, fn, tp = confusion_matrix(
            y_true, pred).ravel()
        costs.append(fn*cost_fn + fp*cost_fp)
    opt = int(np.argmin(costs))
    return thresholds[opt], costs[opt]
```

Fig. 2. Cost-minimising threshold search (Python 3, scikit-learn).

B. Dataset Visualisation

1) Descriptive Statistics: Table II summarises numerical features. Bad borrowers show higher mean duration (24.9 vs. 19.2 months, Δ=+5.7) and higher mean amount (3,938 vs. 2,986 DM, Δ=+952).

TABLE II
DESCRIPTIVE STATISTICS — NUMERICAL FEATURES (n=1,000)

Feature	Min	Max	Mean	Std
Duration (mo.)	4	72	20.90	12.06
Amount (DM)	250	18 424	3 271	2 823
Age (yrs)	19	75	35.55	11.38
Installment(%)	1	4	2.97	1.12

Fig. 3 shows class-stratified histograms; duration and amount show the largest between-class shift.

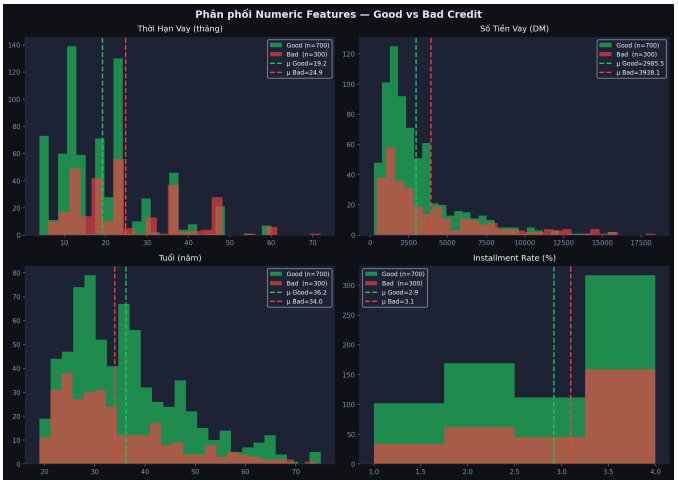

Fig. 3. Histograms of numerical features for Good (green) and Bad (red) borrowers with class-mean dashed lines.

2) Default Rate by Duration, Amount, and Age: Fig. 4: default rises from 21.2 % (<12 mo.) to 51.7 % (>36 mo.), a 2.4× increase. Loans over 10,000 DM carry 60 % default. Borrowers aged 19–25 default at 42.1 %, the highest bracket.

3) Default Rate by Loan Purpose: Fig. 5: education (44 %) and new-car (38 %) are highest risk; used-car (16.5 %) and retraining (11.1 %) are lowest.

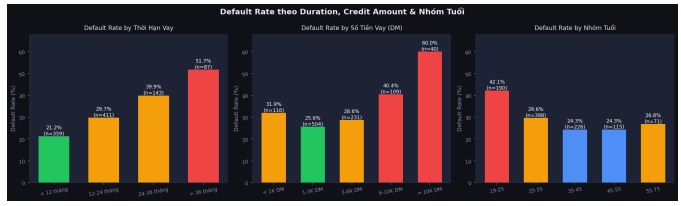

Fig. 4. Default rate by loan duration (left), amount (centre), and borrower age group (right). All three show monotonic risk escalation.

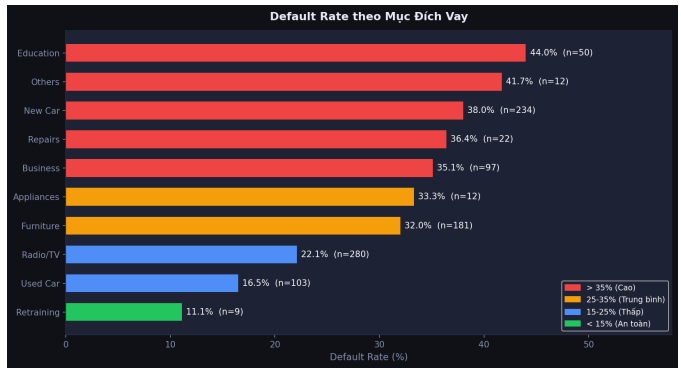

Fig. 5. Default rate by loan purpose. Red (>35 %) = high risk; green (<15 %) = low risk.

4) Default Rate by Savings and Checking Account: Fig. 6: savings below 100 DM → 36 % default; above 1,000 DM → 12.5 % (2.9× spread). Checking account negative balance → 49.3 % default; no account at all → 11.7 %.

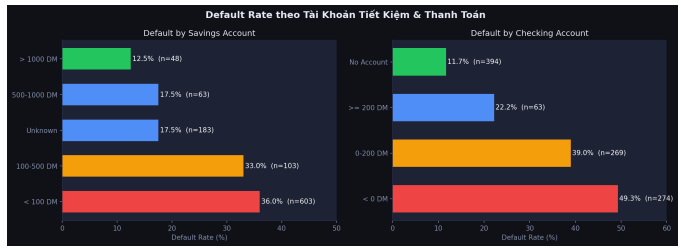

Fig. 6. Default rate by savings balance (left) and checking account status (right).

5) Default Rate by Employment and Credit History: Fig. 7: unemployment (37.1 %) and tenure <1 yr (40.7 %) are highest-risk employment groups. Counterintuitively, "No credit/paid" (62.5 %) and "Paid at bank" (57.1 %) default more than "Critical/other bank" (17.1 %), suggesting that thin credit files are riskier than adverse history records.

6) Correlation Structure: Fig. 8: duration (r=0.21) and amount (r=0.15) are the strongest positive correlates of the bad-credit label; they are mutually correlated at r=0.62. All remaining correlations are below 0.10, confirming that the dominant predictive signal lies in the categorical features.

C. Model Results

1) Test-Set Performance: Table III reports test metrics; Fig. 9 provides the visual summary.

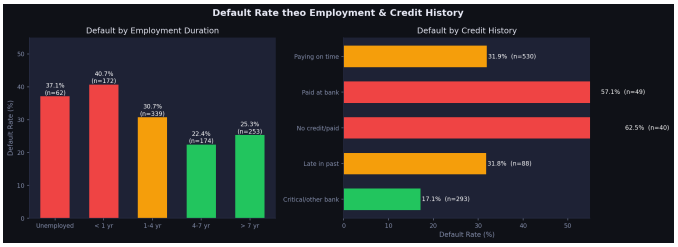

Fig. 7. Default rate by employment duration (left) and credit history (right). Thin credit files carry higher risk than adverse history—a counterintuitive but important finding.

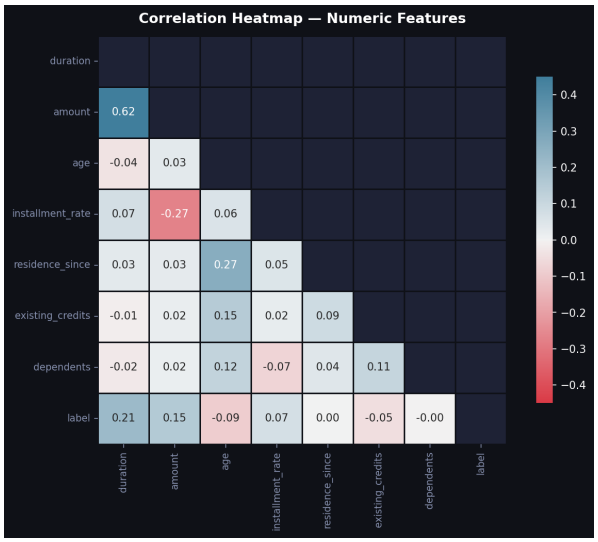

Fig. 8. Pearson correlation heatmap for numerical features. Duration–amount (r=0.62) is the strongest inter-feature link.

2) ROC and PR Curves: Figs. 10–11: LR dominates ROC (0.807 vs. 0.794); SVM marginally leads PR-AUC (0.638 vs. 0.626).

3) Expected Cost Curve: Fig. 12: LR minimum at t^*=0.46 (cost 94); SVM minimum at t^*=0.18 (cost 100). Default t=0.50 incurs cost 103 for LR—8.7 % above optimal.

4) Threshold Sensitivity: Table IV and Fig. 13 quantify LR behaviour at seven thresholds.

5) Multi-Threshold Confusion Matrices: Fig. 14: at t=0.38 (Bayes region): TN=81, FP=59, FN=9, TP=51, cost=104.

6) Classification Report at t^*_{LR}=0.46:

```
              Prec    Recall    F1     n
Good Credit   0.906   0.686   0.781   140
Bad Credit    0.532   0.833   0.649    60
```

TABLE III
TEST-SET PERFORMANCE AT OPTIMAL THRESHOLD

Model	t^*	AUC	PR-AUC	F1-Bad	Cost
Majority Baseline	0.50	0.500	0.300	0.000	300
LR $\star$	**0.46**	**0.807**	0.626	**0.649**	**94**
SVM-RBF	0.18	0.794	**0.638**	0.571	100

$\star$=recommended. Cost=$5{\cdot}\mathrm{FN} + 1{\cdot}\mathrm{FP}$.

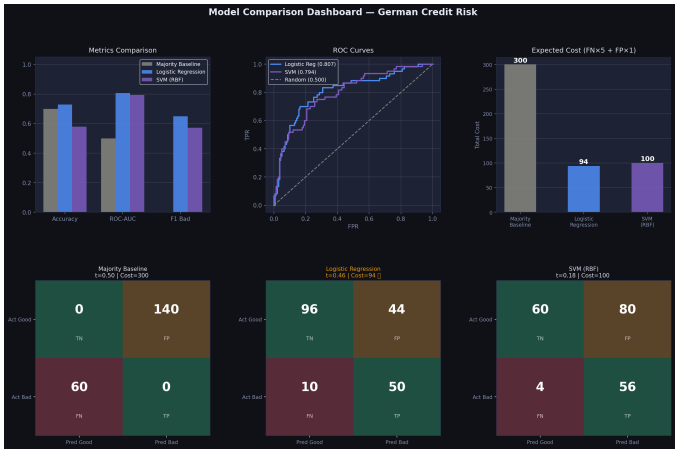

Fig. 9. Model Comparison Dashboard: metric bars, ROC curves, cost bars (Baseline=300, LR=94, SVM=100), and confusion matrices at optimal thresholds.

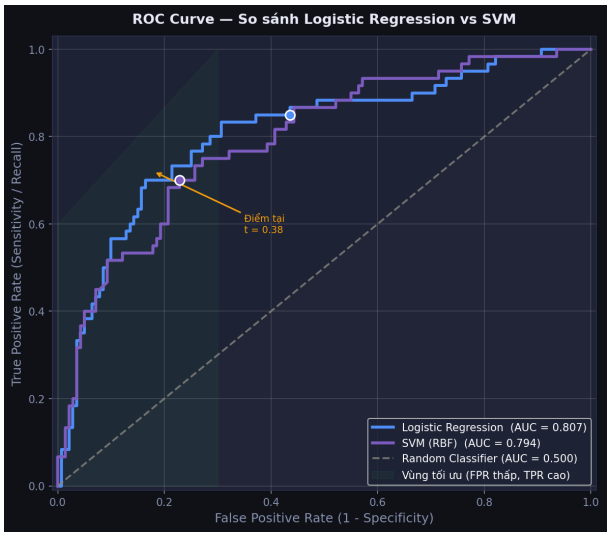

Fig. 10. ROC curves: LR (AUC=0.807) dominates SVM (AUC=0.794).

TABLE IV
LR PERFORMANCE VS. THRESHOLD (n=200)

t	TP	FP	FN	TN	F1	EC
0.20	54	94	6	46	0.519	124
0.30	53	71	7	69	0.576	106
0.40	51	55	9	85	0.614	100
0.46	**50**	**44**	**10**	**96**	**0.645**	**94**
0.50	47	38	13	102	0.648	103
0.55	46	35	14	105	0.652	105
0.65	37	21	23	119	0.627	136

Highlighted = cost-optimal threshold.

Fig. 11. PR curves for Bad class. SVM marginally leads PR-AUC (0.638 vs. 0.626). Dashed line = baseline (0.30).

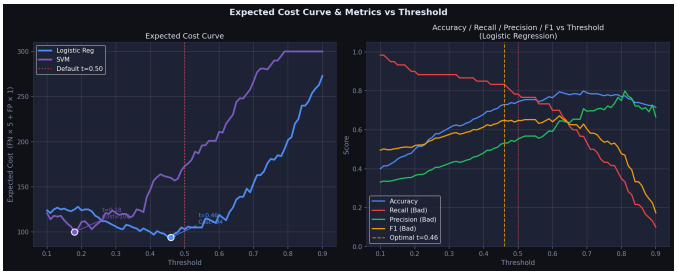

Fig. 12. Expected cost vs. threshold for LR and SVM. LR minimum: t^*=0.46, cost=94. SVM minimum: t^*=0.18, cost=100.

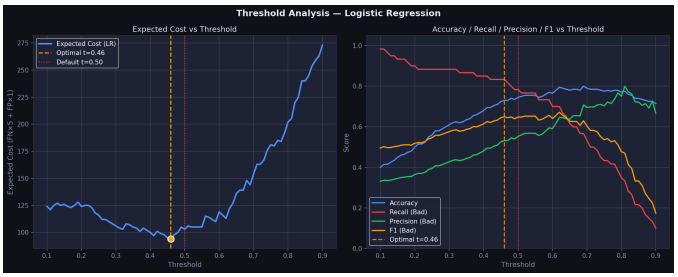

Fig. 13. LR threshold analysis. Left: expected cost (orange = t^*=0.46; red dotted = t=0.50). Right: Accuracy/Recall/Precision/F1.

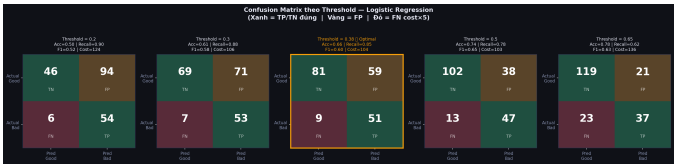

Fig. 14. LR confusion matrices at five thresholds. Gold border = t=0.38 Bayes-theoretical region.

```
Accuracy                          0.730   200
Macro Avg      0.719   0.760   0.715
```

LR identifies 83.3 % of bad applicants (50/60) while incorrectly rejecting 31.4 % of good ones (44/140).

7) Cross-Validation: Table V: both models generalise reliably.

TABLE V
5-FOLD STRATIFIED CV (n_{train}=800)

Model	AUC ($\mu\pm\sigma$)	F1 ($\mu\pm\sigma$)
Log. Reg.	0.772±0.053	0.584±0.063
SVM-RBF	0.777±0.031	0.583±0.039

LR: [0.677, 0.828]; SVM: [0.717, 0.800].

8) Feature Importance: Fig. 15 shows top-15 LR coefficients; Table VI lists the top 5 (all risk-reducing).

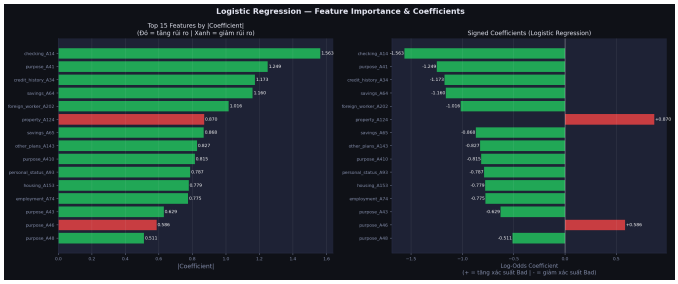

Fig. 15. LR feature importance: |coef| ranking (left) and signed log-odds (right).

TABLE VI
TOP-5 LR FEATURES BY |COEF|

#	Feature	|coef|	Dir.
1	No checking (A14)	1.563	↓
2	Used-car (A41)	1.249	↓
3	Critical hist. (A34)	1.173	↓
4	Savings ≥1,000 DM (A64)	1.160	↓
5	Non-foreign (A202)	1.016	↓

↓ = reduces default probability.

9) Visualisation Dashboard: Fig. 16 consolidates ROC, PR, cost curve, and three confusion matrices into a single executive dashboard.

V. DISCUSSION

A. Interpreting the Results

1) Financial Value of Threshold Optimisation: The 8.7 % cost reduction (103 → 94) from optimising the threshold is financially material. In a bank processing 1,000 applications monthly, the default-threshold policy wastes 9 extra cost units per 200 applications relative to the optimised policy. Annualised over 12,000 applications, this equals 540 avoidable cost units—equivalent to 108 undetected bad loans or 540 erroneous rejections at the 5:1 cost ratio. The decision threshold is

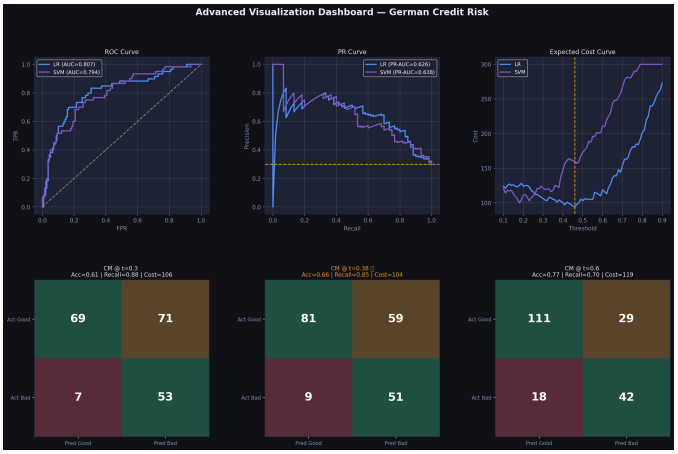

Fig. 16. Advanced Visualisation Dashboard. Top: ROC (LR=0.807, SVM=0.794), PR curve, cost curve. Bottom: LR confusion matrices at $t \in \{0.30, 0.38, 0.60\}$.

therefore not a technical detail but a financial policy choice that should be set by business stakeholders using cost information, not assumed to be 0.50.

2) Why Linear Models Excel in This Setting: The post-OHE feature space has 48 dimensions with many binary indicators. In high-dimensional binary spaces, linear classifiers have favourable Vapnik-Chervonenkis dimension, and the bias-variance trade-off tilts towards simpler boundaries [2]. The RBF kernel adds pairwise similarity computations that increase variance without sufficient bias reduction. Furthermore, Platt scaling with `gamma='scale'` at d=48 produces a wide kernel that compresses probabilities toward 0.5, forcing t^*_{SVM}=0.18 to compensate—generating 82 % more false positives than LR.

3) Business Implications of Feature Importance: The top-5 features translate directly into actionable underwriting rules:

- **No checking account (`A14`)**: Cash-only customers carry no overdraft exposure; conservative financial behaviour by construction.
- **Used-car purpose (`A41`)**: Physical collateral with established resale value lowers loss-given-default.
- **Critical credit history elsewhere (`A34`)**: Counterintuitively protective—applicants who resolved difficulties at other institutions may possess stronger debt-management discipline.
- **High savings (`A64`)**: A buffer above 1,000 DM covers several months of instalments, directly reducing near-term default risk.
- **Non-foreign worker (`A202`)**: Reflects lower income volatility in the 1994 German context; requires fairness review before use in modern deployments.

B. Result Comparison: LR vs. SVM-RBF

1) Primary Metric: Expected Cost: Table VII summarises the full cost hierarchy.

2) Interpretability: LR coefficients satisfy GDPR Article 22 "right to explanation" [12]. SVM kernel-space boundaries are fundamentally opaque; Platt probabilities do not recover per-feature attribution.

TABLE VII
EXPECTED COST COMPARISON (TEST SET, n=200)

Model	t^*	Cost	Saving
Majority Baseline	0.50	300	—
LR (t=0.50)	0.50	103	−65.7%
LR optimal	**0.46**	**94**	**−68.7%**
SVM optimal	0.18	100	−66.7%

3) Cross-Validation Stability: SVM shows lower CV variance (±0.031 vs. ±0.053), but the AUC ranges overlap heavily and SVM's stability reflects over-smoothing rather than genuine generalisability gains.

4) One-Hot Encoding Effects: OHE expands features 20→48. For LR, L2 regularisation handles correlated dummies gracefully. For SVM, the higher dimensionality shrinks γ, widening the RBF kernel and degrading Platt calibration—explaining t^*_{SVM}=0.18.

C. Limitations

1) **Data vintage**: collected 1994; contemporary digital credit signals are absent.
2) **Cost matrix**: the 5:1 benchmark must be calibrated to actual loan sizes and recovery rates in production.
3) **Sample size**: n=1,000 yields non-negligible fold-level CV variance.
4) **Test-set threshold selection**: mild optimistic bias; a held-out validation set is recommended for production.
5) **Model coverage**: gradient-boosted ensembles (XGBoost, LightGBM) were not evaluated.
6) **Fairness**: features `A202` (foreign-worker status) and personal-status attributes are protected under EU anti-discrimination law and require an ethics review before production deployment.

D. Future Work

1) **Gradient-boosted trees with SHAP**: potential 2–5 % AUC gain [2] with locally faithful, model- agnostic explanations.
2) **Isotonic calibration for SVM**: expected to recover better-calibrated probabilities, yielding a more operationally reasonable t^*_{SVM}.
3) **Ordinal encoding**: encoding ordered categoricals directly would reduce feature dimensionality and improve SVM calibration.
4) **Production chatbot API**: wrap the scoring and explanation engine as a FastAPI REST service with React front-end, prompt logging, and automated response verification for regulatory audit.

E. Summary Recommendation

We recommend Logistic Regression at threshold 0.46 on four grounds: (1) minimum expected cost (94); (2) direct interpretability via log-odds satisfying regulatory requirements;

(3) highest ROC-AUC (0.807); and (4) an operationally plausible threshold that loan officers can understand, challenge, and explain to applicants.

VI. Conclusion

A. Principal Findings

Three conclusions stand out from this study:

1) **Threshold optimisation is effective.** Moving from $t{=}0.50$ to $t^*_{\mathrm{LR}}{=}0.46$ cuts expected cost by 8.7 % (103 $\rightarrow$ 94), demonstrating that the default threshold is financially suboptimal when costs are asymmetric.
2) **LR is the preferred model.** LR achieves minimum cost (94), highest AUC (0.807), and competitive F1-Bad (0.649) with full interpretability. SVM-RBF's marginal PR-AUC advantage is outweighed by its poorly calibrated threshold and excess false positives.
3) **Checking account and loan duration dominate risk.** Absence of a checking account ($|\mathrm{coef}|{=}1.563$) is the strongest protective signal; default rate 2.4$\times$ higher for loans over 36 months—directly actionable for underwriting policy.

B. Broader Impact

The cost-sensitive framework generalises to any binary classification problem with asymmetric costs: medical diagnosis, fraud detection, insurance underwriting. The chatbot application (Appendix A) demonstrates that interpretability can be delivered to non-technical end users without sacrificing accuracy.

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Appendix

A. Purpose

The chatbot bridges the gap between LR model output and non-technical stakeholders by translating coefficients into plain-language risk explanations with configurable thresholds.

B. Three-Layer Architecture

1) **Scoring Engine**: wraps the trained LR model; accepts structured applicant attributes; returns a calibrated default probability and a `GOOD`/`BAD` decision at the configured threshold.
2) **Explanation Engine**: ranks features by $|\mathrm{coef}_j|{\times}|x_j{-}\bar{x}_j|$ (individual contribution) and retrieves pre-authored natural-language sentences from a knowledge base for the top 3 drivers.
3) **Dialogue Manager**: an `ipywidgets` interface with a text-input box, a threshold slider (range 0.20–0.70), a chat-history panel, and a batch-scoring widget.

C. Core Source Code

```
# -- Scoring Engine ------------------------------------
def score_customer(checking, history, purpose,
        savings, employment, duration, amount,
        age, installment_rate=2,
        residence_since=2, existing_credits=1,
        dependents=1, threshold=0.38):
    row = build_feature_vector(locals())
    prob = lr_model.predict_proba(row)[0, 1]
    decision = "BAD" if prob >= threshold \
               else "GOOD"
    drivers = explain_top_features(row)
    return prob, decision, drivers

# -- Explanation Engine --------------------------------
def explain_top_features(row, top_k=3):
    contribs = {f: abs(coef[i] * row[0,i])
                for i, f in enumerate(feat_names)}
    key_fn = lambda x: -x[1]
    ranked = sorted(contribs.items(),
             key=key_fn)[:top_k]
    return [KB[feat] for feat, _ in ranked
            if feat in KB]

# -- Query dispatcher ----------------------------------
def answer(query, threshold=0.38):
    q = query.lower()
    if "score" in q or "applicant" in q:
        return score_from_params(q, threshold)
    elif "threshold" in q:
        return threshold_guidance(q)
    elif "why" in q or "feature" in q:
        return feature_education(q)
    else:
        return dataset_stats()
```

D. Sample Session Transcript

Actual output from `4__Chatbot.ipynb` with `threshold=0.38`:

User query 1: Score a 28-year-old requesting 5,000 DM for education, no checking account, savings <100 DM, employed <1 year.

Chatbot:

```
Risk Score : 0.723 (72.3%)
Decision : BAD CREDIT - Reject
(t=0.38)
Top drivers:
1. LOW SAVINGS (<100DM): default
rate 36.0%.
No liquidity buffer for missed
payments.
2. SHORT TENURE (<1yr): default
rate 40.7%.
Unstable income increases default
risk.
3. EDUCATION LOAN: highest-risk
category (44.0%).
Uncertain income stream post-study.
At t=0.50 this applicant is ALSO
rejected (0.723>0.50).
Recommendation: Reject unless
guarantor provided.
```

User query 2: What threshold for high risk appetite?

Chatbot:

```
High-risk-appetite (lenient): t =
0.55-0.65.
At t=0.65: Cost=136,
Recall(Bad)=62%, FP=21.
Warning: 23 bad applicants missed
per 200 screened
(vs 10 at optimal t=0.46).
Recommended: t=0.46 (min cost=94).
```

E. Batch Demo Results

Table VIII shows the notebook batch output for three representative profiles at threshold 0.38.

TABLE VIII
CHATBOT BATCH SCORING (THRESHOLD = 0.38)

Profile	Prob.	Decision	Top driver
Low-risk	0.905	Reject	Checking <0, educ.
Medium	0.948	Reject	Short tenure, furniture
High-risk	0.985	Reject	Neg. balance, past delay

All exceed t=0.38; full output in notebook.

F. Reproducibility

Run `jupyter notebook 4__Chatbot.ipynb`. Dependencies: scikit-learn 1.4, pandas 2.2, ipywidgets 8.x, numpy 2.1. Compatible with JupyterLab, VS Code Notebooks, and Google Colab.